

SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-016: Show / Hide Password Visibility Toggle

Field	Details
Change Request ID	CR-016
Change Request Title	Show / Hide Password Visibility Toggle
Student Name	C. Lavasujan
Student ID	24ICT06
Implementation Status	Completed
Date	09 October 2026

1. Change Request Summary

1.1 Business Requirement

Users and administrators previously entered passwords into masked input fields without an option to view or verify the typed text. Because passwords can contain numbers, symbols, and mixed letter casing, typographical errors can cause avoidable failed login attempts. CR-016 adds a simple client-side visibility toggle so users can check their entry before submitting credentials.

1.2 Existing Behaviour

Before implementation:

- `LoginFormSection.jsx` rendered the username and password fields for the login flow.
- The password input used `type="password"` and characters remained masked at all times.
- No eye / eye-slash control was available to switch the password display.
- The existing login request submitted credentials through the established authentication flow.
- `StudentFormModal.jsx` did not have the same visibility toggle in the Admin Dashboard Add Student modal.

1.3 Requested Change

Implement a show/hide control for the login password field and extend the same user-facing capability to the Add Student modal. The requested behaviour includes:

- Switch the input type between `password` and `text` through component state.
- Keep the password masked by default whenever the component loads.
- Show an eye icon while hidden and an eye-slash icon while visible.
- Expose dynamic `aria-label` and `aria-pressed` attributes for accessible state feedback.
- Use `type="button"` so the toggle itself cannot submit the form.
- Add spacing and positioning so entered text does not overlap the icon.
- Support keyboard activation using Tab, Space, and Enter.
- Preserve existing authentication, validation alerts, and role-based navigation.

2. Implementation

2.1 Implementation Summary

The password toggle was implemented in the login component and extended to the Admin Dashboard Add Student modal. The React state `showPassword` controls whether the password input is masked or displayed as text. Eye and EyeOff SVG icons communicate the current state, while accessibility attributes describe the action to assistive technology. CSS positions the toggle inside the field boundary and provides right padding so the password remains legible.

2.2 Affected Files

File	Change	Purpose
<code>frontend/src/components/LoginFormSection.jsx</code>	Modified	Added <code>showPassword</code> state, dynamic input type, accessible toggle button, and icon updates.
<code>frontend/src/styles/login.css</code>	Modified	Added password input wrapper and toggle-button positioning / hover styles.
<code>frontend/src/components/StudentFormModal.jsx</code>	Modified	Extended the visibility toggle to the Admin Dashboard Add Student password field.
<code>frontend/src/styles/dashboard.css</code>	Modified	Styled the password wrapper and toggle within the student modal.

2.3 Technical Changes

Visibility State Management

A local React state variable manages the visibility state:

- `const [showPassword, setShowPassword] = useState(false);` initializes the input in the masked state.
- When `showPassword` is false, the input type is `password`, the accessible label prompts “Show password”, and the Eye icon is displayed.
- When `showPassword` is true, the input type becomes `text`, the label changes to “Hide password”, and the EyeOff icon is displayed.

Form Submission and Layout

- The toggle is a native button with `type="button"`, preventing accidental login submission when pressed.
- The password input uses right padding (44px) so text does not collide with the absolutely positioned toggle.
- The same interaction and corresponding styling are available in the Admin Dashboard Add Student modal.

Accessibility and Security

- The button is reachable using Tab and activated using Space or Enter.
- Dynamic `aria-label` and `aria-pressed` values expose the current state to assistive technology.
- The password remains masked on initial load and is not persisted in `localStorage` or `sessionStorage`.
- Plaintext is displayed only after explicit user interaction; users should re-mask the field when finished in shared environments.

2.4 Implementation Logic

1. The component mounts with `showPassword` set to false, so password characters are hidden.
2. The user enters a password in the input field.
3. The user clicks the visibility button or focuses it and presses Space / Enter.
4. The state changes and the input type switches between `password` and `text`.
5. The icon and accessible label update to represent the new visibility state.
6. Padding preserves legibility and prevents the typed characters from being hidden behind the button.
7. The user can activate the button again to mask the password.
8. Form submission continues to pass the entered credentials to the existing login function unchanged.

2.5 Error and Data Handling

- Empty username or password continues to display the existing required-field validation message.
- Invalid credentials continue to display the existing authentication error banner.
- Valid credentials can be submitted whether the password is masked or temporarily visible.
- The toggle does not change the credential value; it changes only the input presentation state.

3. Testing & Verification

3.1 Testing Approach

The implementation was verified through manual frontend and browser testing. The test scope includes functional, positive, UI, accessibility / keyboard, authentication, and regression checks. Environment: Development. Browser: not specified in the supplied implementation note. Test context: Login module and Admin Dashboard Add Student modal.

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Default masked state on load	Password characters are masked by default on page load.	Input remains masked when the component mounts.	Pass
TC-02	Click the visibility toggle	Password changes to visible text; clicking again masks it.	Toggle changes visibility correctly in both directions.	Pass
TC-03	Dynamic icon updates	Eye icon is shown when masked and EyeOff when visible.	Icon updates with the visibility state.	Pass
TC-04	Toggle does not submit the form	Clicking the toggle does not trigger premature form submission.	Form is not submitted by the toggle.	Pass
TC-05	Keyboard accessibility	Toggle is reachable with Tab and activated with Space / Enter.	Keyboard focus and activation work.	Pass
TC-06	Authentication with visible password	Valid credentials authenticate while the password is shown as text.	Login succeeds with visible credentials.	Pass
TC-07	Admin Add Student modal	Administrator can toggle visibility in the Add Student modal.	Toggle works in the Admin Dashboard modal.	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	7
Passed	7
Failed	0
Blocked	0
Pass Rate	100%
Overall Test Result	Passed

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification	Status
AC-001	Users can toggle password visibility between hidden and visible.	Functional testing (TC-02, TC-03)	Met
AC-002	Password field is masked by default on page load.	Initial-state check (TC-01)	Met
AC-003	Toggle icon updates to match visibility state.	Icon update check (TC-02, TC-03)	Met
AC-004	Using the toggle does not submit the login form.	Form behaviour check (TC-05)	Met
AC-005	Toggle is keyboard accessible and has an accessible label.	Keyboard / accessibility check (TC-06)	Met
AC-006	Password text is not obscured by the toggle icon.	Layout / padding check (TC-04)	Met
AC-007	Valid credentials work while password is visible or masked.	Login regression check documented in source report	Met
AC-008	Existing validation messages and role-based redirects continue to work.	Authentication / regression check documented in source report	Met

5. Evidence

Insert the three captured screenshots below. The screenshot placeholders are editable and should be replaced with the original screenshots before submission.

Evidence 01 - Login page with masked password

Related: AC-001, AC-002, AC-003, AC-005, AC-006

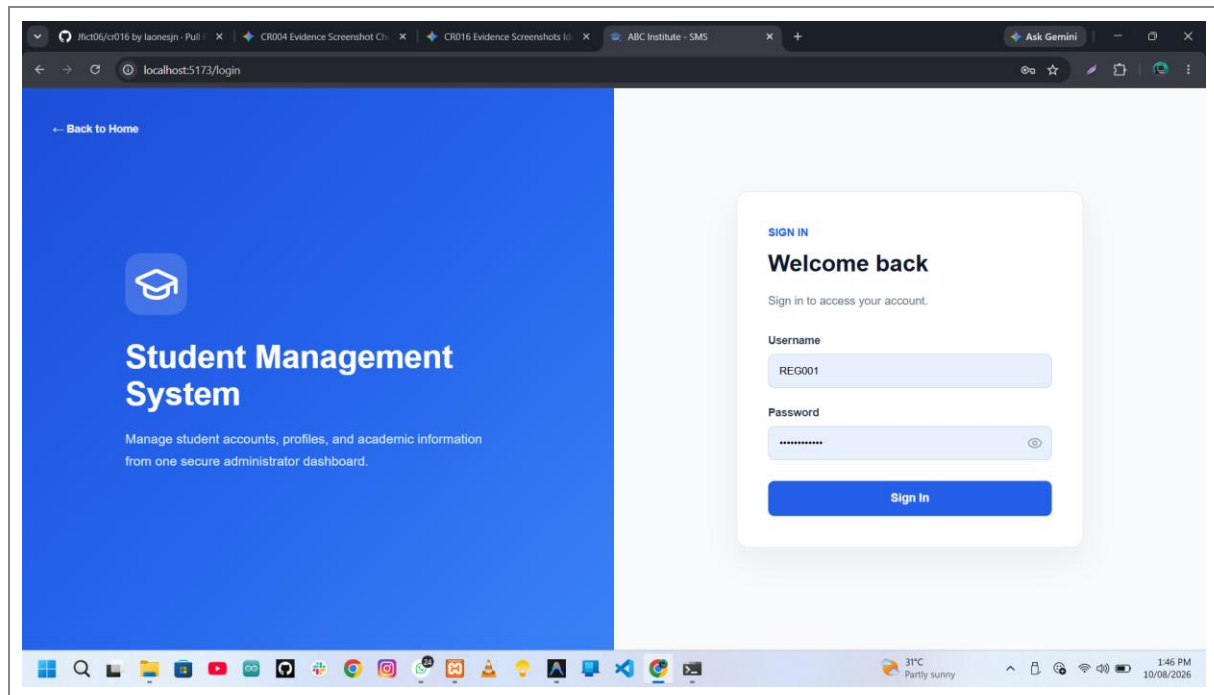


Figure 1: Login page showing the default masked password state and eye toggle.

Evidence 02 - Login page with visible password

Related: AC-001, AC-003, AC-004, AC-007, AC-008

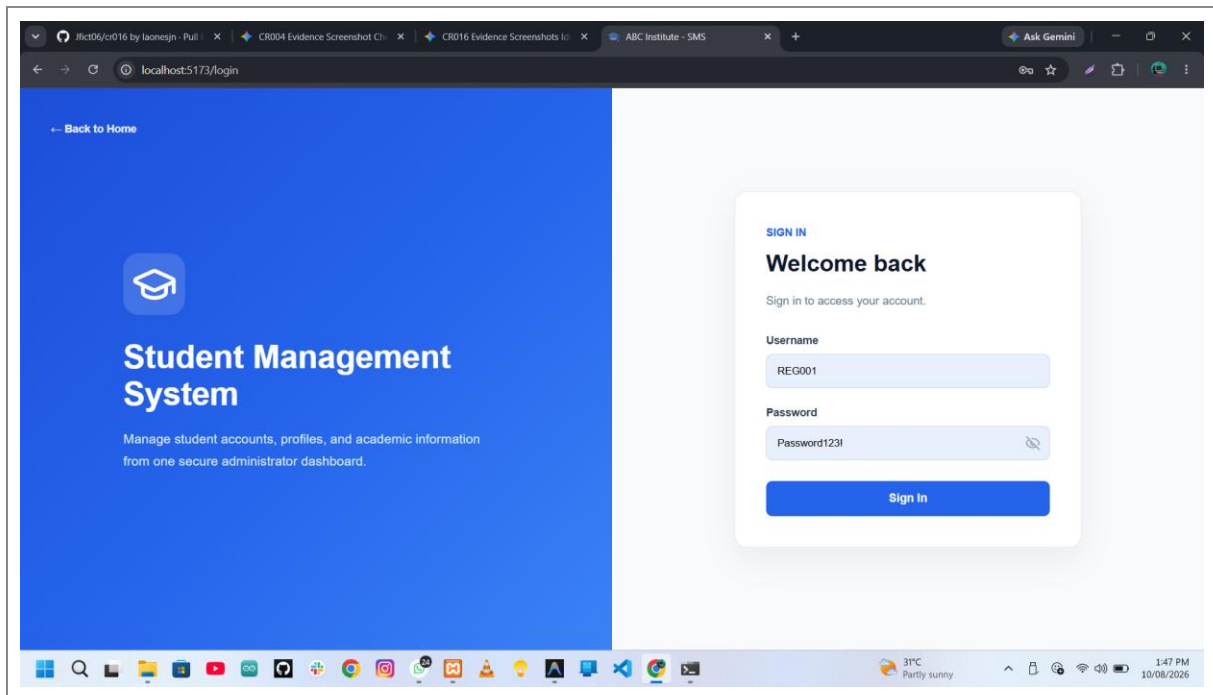


Figure 2: Login page showing the password text after activating the visibility toggle.

Evidence 03 - Admin Dashboard Add Student modal

Related: extended CR-016 requirement; TC-07

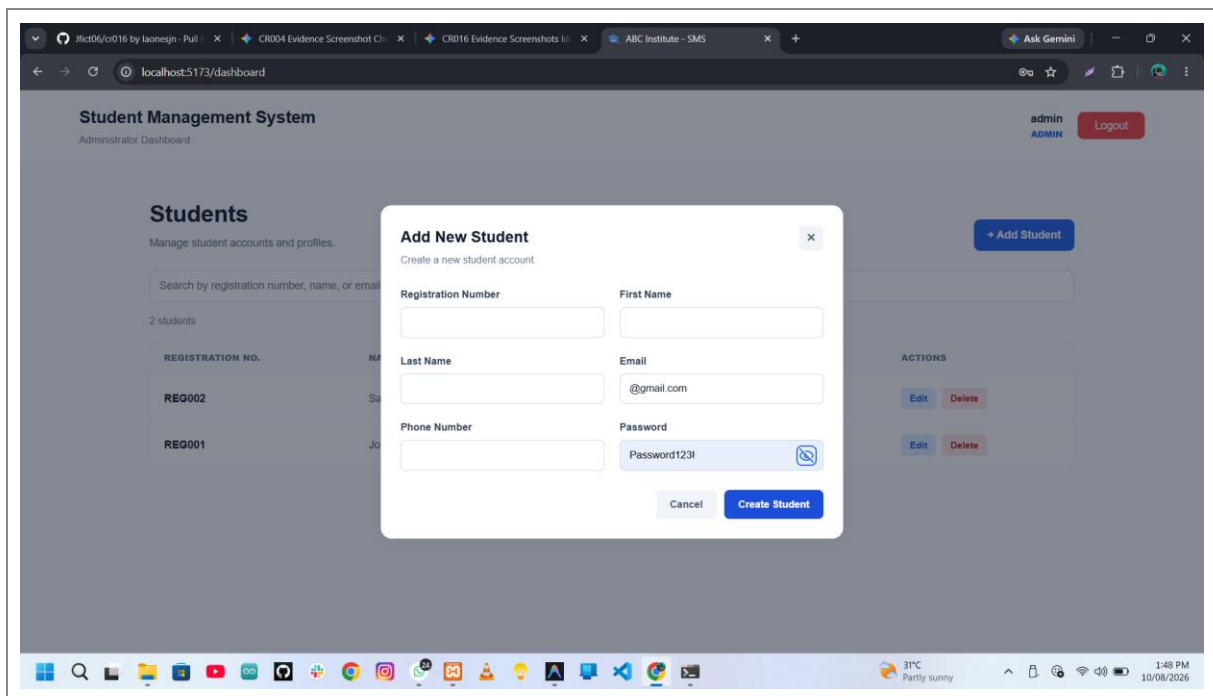


Figure 3: Admin Dashboard Add Student modal with password visibility enabled.

6. Final Result

Item	Result
Requirements	Met
Implementation	Completed
Testing	Passed - 7/7 test cases
Acceptance Criteria	All 8 criteria met
Existing Authentication	Preserved / verified in supplied implementation note
Final Status	Completed - screenshot insertion pending

6.1 Final Summary

CR-016 implements a client-side show / hide password visibility toggle in the login module and extends the same behaviour to the Admin Dashboard Add Student modal. The user can view or mask entered text on demand, with the masked state as the default. The toggle uses React state, eye / eye-slash indicators, accessible labels, keyboard activation, and layout padding that keeps the password readable. Existing authentication and validation flows remain unchanged. The supplied implementation note records all seven test cases as passed.

6.2 Issues / Limitations

No functional issues are identified in the supplied implementation summary. The three requested screenshots still need to be inserted into the evidence placeholders because the screenshot image files were not included with the Markdown source.

Student Name: C. Lavasujan

Student ID: 24ICT06

Date: 09 October 2026

Signature: _____

~ END OF REPORT ~